Unit-I
Introduction to Cloud Computing Fundamentals

FUNDAMENTALS OF CLOUD COMPUTING

Unit -I

- Introduction to Cloud Computing Fundamentals
- Cloud Computing at a Glance,
- Historical Developments,
- Building Cloud Computing Environments,
- Computing Platforms and Technologies.

Concepts

- Introduction
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- Building Cloud Computing Environments,
  - Application Development
  - Infrastructure and System Development
- Computing Platforms and Technologies,
  - Amazon Web Services (AWS)
  - Google AppEngine,
  - Microsoft Azure,
  - Hadoop
  - Force.com and Salesforce.com,
  - Manjrasoft Aneka

Introduction Cloud Computing

- Cloud computing is a virtualization-based technology that allows us to create, configure, and customize applications via an internet connection.
- The cloud technology includes a development platform, hard disk, software application, and database.

What is Cloud Computing

- The term cloud refers to a network or the internet. It is a technology that uses remote servers on the internet to store, manage, and access data online rather than local drives.
- The data can be anything such as files, images, documents, audio, video, and more.

Cloud Computing operation

- There are the following operations that we can do using cloud computing:
  - Developing new applications and services
  - Storage, back up, and recovery of data
  - Hosting blogs and websites
  - Delivery of software on demand
  - Analysis of data
  - Streaming videos and audios

Why Cloud Computing?

- Small as well as large IT companies, follow the traditional methods to provide the IT infrastructure. That means **for any IT company, we need a Server Room that is the basic need of IT companies**.
- In that server room, there should be a database server, mail server, networking, firewalls, routers, modem, switches, QPS (Query Per Second means how much queries or load will be handled by the server), configurable system, high net speed, and the maintenance engineers.
- To establish such IT infrastructure, we need to spend lots of money. To overcome all these problems and to reduce the IT infrastructure cost, Cloud Computing comes into existence.

Cloud Computing at a Glance

- In 1969, Leonard Kleinrock, one of the chief scientists of the original Advanced Research Projects Agency Network (ARPANET) which seeded the Internet, said:
- *"As of now, computer networks are still in their infancy, but as they grow up and become sophisticated, we will probably see the spread of 'computer utilities' which, like present electric and telephone utilities, will service individual homes and offices across the country."*

Cloud Computing at a Glance

- Cloud computing helps users in easily accessing computing resources like storage, and processing over internet rather than local hardware's. Here we discussing how it works in nutshell:
- **Infrastructure:** Cloud computing depends on remote network servers hosted on internet for store, manage, and process the data.
- **On-Demand Access:** Users can access cloud services and resources based on-demand they can scale up or down the without having to invest for physical hardware.
- **Types of Services:** Cloud computing offers various benefits such as cost saving, scalability, reliability and acessibility it reduces capital expenditures, improves efficiency.

Cloud Computing at a Glance

- Nowadays, **Cloud computing** is adopted by every company, whether it is an MNC or a startup many are still migrating towards it because of the cost-cutting, lesser maintenance, and the increased capacity of the data with the help of servers maintained by the cloud providers.
- One more reason for this drastic change from the On-premises servers of the companies to the Cloud providers is the **'Pay as you go**' principle-based services provided by them i.e., you only have to pay for the service which you are using. The disadvantage On-premises server holds is that if the server is not in use the company still has to pay for it.

Cloud Computing at a Glance

- One of the most diffused views of Cloud computing can be summarized as follows:
- *“I don’t care where my servers are, who manages them, where my documents are stored, or where my applications are hosted. I just want them always available and access them from any device con- nected through Internet. And I am willing to pay for this service for as a long as I need it.”*

The Vision (Aims to be) of Cloud Computing

- The vision of cloud computing is to make data, applications, and services accessible over the internet instead of on local devices or servers.
- Cloud computing allows anyone having a credit card to provision **virtual hardware, runtime environments, and services.**
- These are used for as long as needed and no upfront commitments are required.
- The entire stack of a computing system is transformed into a collection of utilities, which can be provisioned and composed together to deploy systems in hours, rather than days, and with virtually no maintenance costs.

The Vision of Cloud Computing

- These are following **Vision of Cloud Computing :**
- Cloud computing **provides the facility** to provision virtual hardware, runtime environment and services to a person having money.
- These all things can be used as long as they are needed by the user.
- The whole collection of computing system is transformed into **collection of utilities**, which can be provisioned and composed together to deploy systems in hours rather than days, with no maintenance cost.
- The long term vision of a cloud computing is that IT services are traded as utilities in an open market **without technological and legal barriers**.
- In the future, we can imagine that it will be possible to **find the solution that matches with our requirements** by simply entering out request in a global digital market that trades with cloud computing services.

The Vision of Cloud Computing

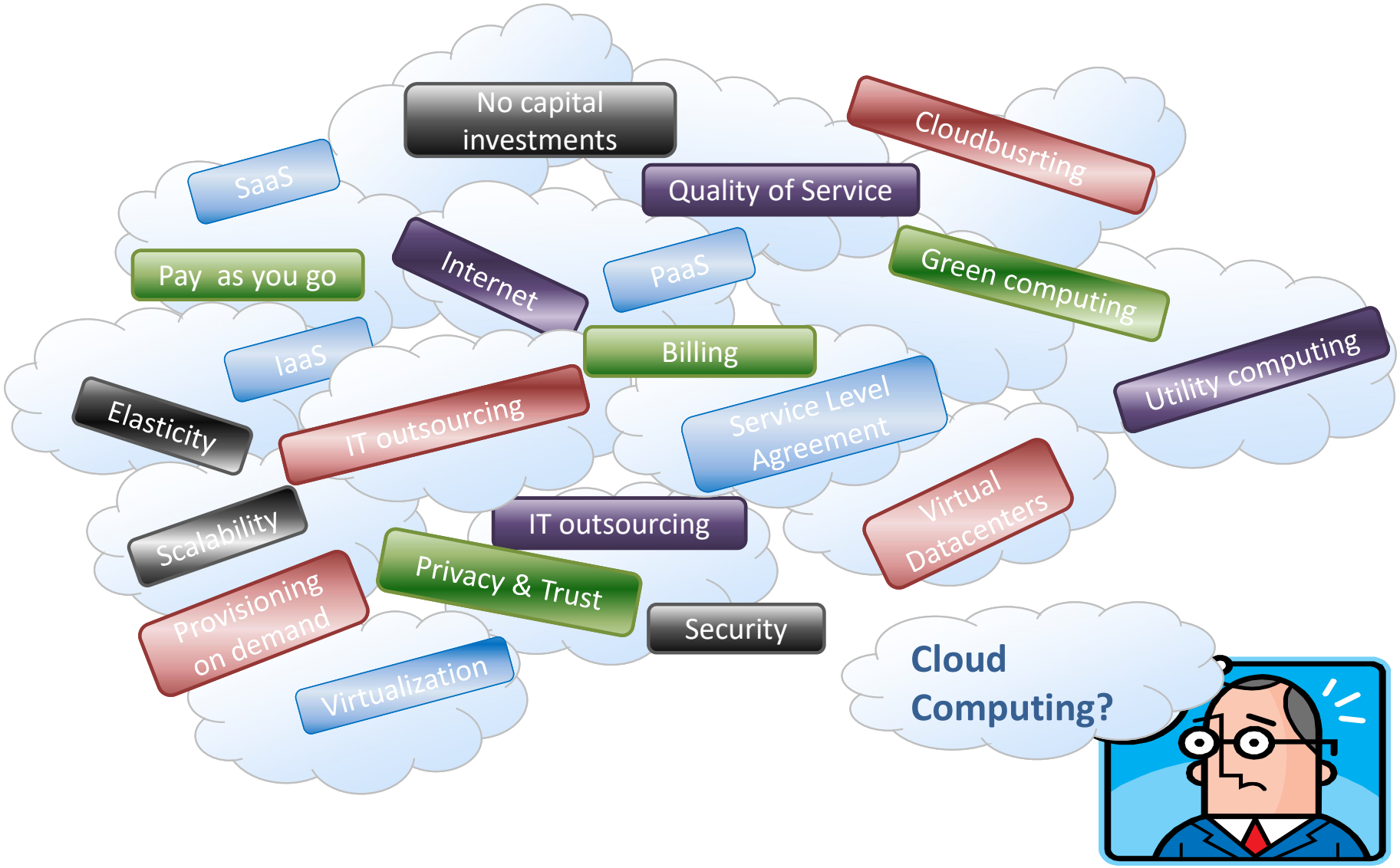
- The existence of such market will enable **the automation of discovery process** and its **integration into** its existing software systems.
- Due to the existence of a **global platform** for trading **cloud services** will also help service providers to potentially increase their revenue.
- A cloud provider can **also become a consumer** of a competition service in order **to fulfil its promises** to customers.
- In the near future we can imagine **a solution that suits our needs** by simply applying our application to the global digital market for cloud computing services.
- The presence of this market will enable the **acquisition process to automatically integrate** with its integration into its existing software applications. The availability of a global cloud trading platform will also help service providers to increase their revenue.
- A cloud provider can also be a buyer of a competitive service to fulfil its promises to customers.

The Vision of Cloud Computing

Defining a Cloud

- Cloud computing has become a **popular buzzword** and it has been widely used to refer to **different technologies, services, and concepts**. It is often associated with **virtualized** infrastructure or hardware on demand, utility computing, IT outsourcing, platform and software as a service, and many other things that now are the focus of the IT industry.
- Figure depicts the plethora(Large amount) of different notions one portrays when defining Cloud computing.

Defining a Cloud



Defining a Cloud

- The term "**Cloud**" has historically been used in the telecommunication industry as an abstraction of the network in system diagrams.
- the definition given by Armbrust et al. [28]:
- *"Cloud computing refers to both the applications delivered as services over the Internet, and the hardware and system software in the datacenters that provide those services."*
- This definition describes Cloud computing as a phenomenon touching on the entire stack: from the underlying hardware to the high level software services and applications

Defining a Cloud

- a definition proposed by American **National Institute of Standards and Technology** (NIST):
- *"Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g. networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction."*

Defining a Cloud

- we can define three criteria to discriminate whether a service is delivered in the Cloud computing style:
  - The service is accessible via a Web browser (non-proprietary) or Web services API.
  - Zero capital expenditure is necessary to get started.
  - You pay only for what you use as you use it.

Defining a Cloud

- The utility-oriented nature of Cloud computing is clearly expressed by Buyya et al. [30] :
- *"A Cloud is a type of parallel and distributed system consisting of a collection of interconnected and virtualized computers that are dynamically provisioned and presented as one or more unified computing resources based on service-level agreements established through negotiation between the service provider and consumers."*

A Closer Look in CC

- Cloud computing is helping enterprises, governments, public and private institutions, as well as re-search organizations shape more effective and demand-driven computing systems.

A Closer Look in CC

- Practical examples of such systems exist across all market segments:
- **1.Large enterprises can offload some of their activities to Cloud- based systems:**
  - Recently, **the New York Times has converted its digital library** about past editions into a Web friendly format.
  - This required a considerable amount of computing power for a **short period** of time.
  - By renting **Amazon EC2 and S3 Cloud resources**, it **performed this task in 36 hours**, and relinquished these resources without any additional costs.

A Closer Look in CC

- **2.Small enterprises and start-ups can afford to translate into business results their ideas more quickly without excessive upfront costs.**
  - **Animoto is a company** that creates videos out of images, music, and video fragments submitted by users. The process involves a considerable amount of storage and backend processing required for producing the video, which is finally made available to the user.
  - Animoto does not own a single server and bases its computing infrastructure entirely on **Amazon Web Services**, which is sized on demand according to the overall workload to be processed. Such workload can vary a lot and requires instant scalability.
  - Upfront investment is clearly not an effective solution and Cloud computing systems become an appropriate alternative.

A Closer Look in CC

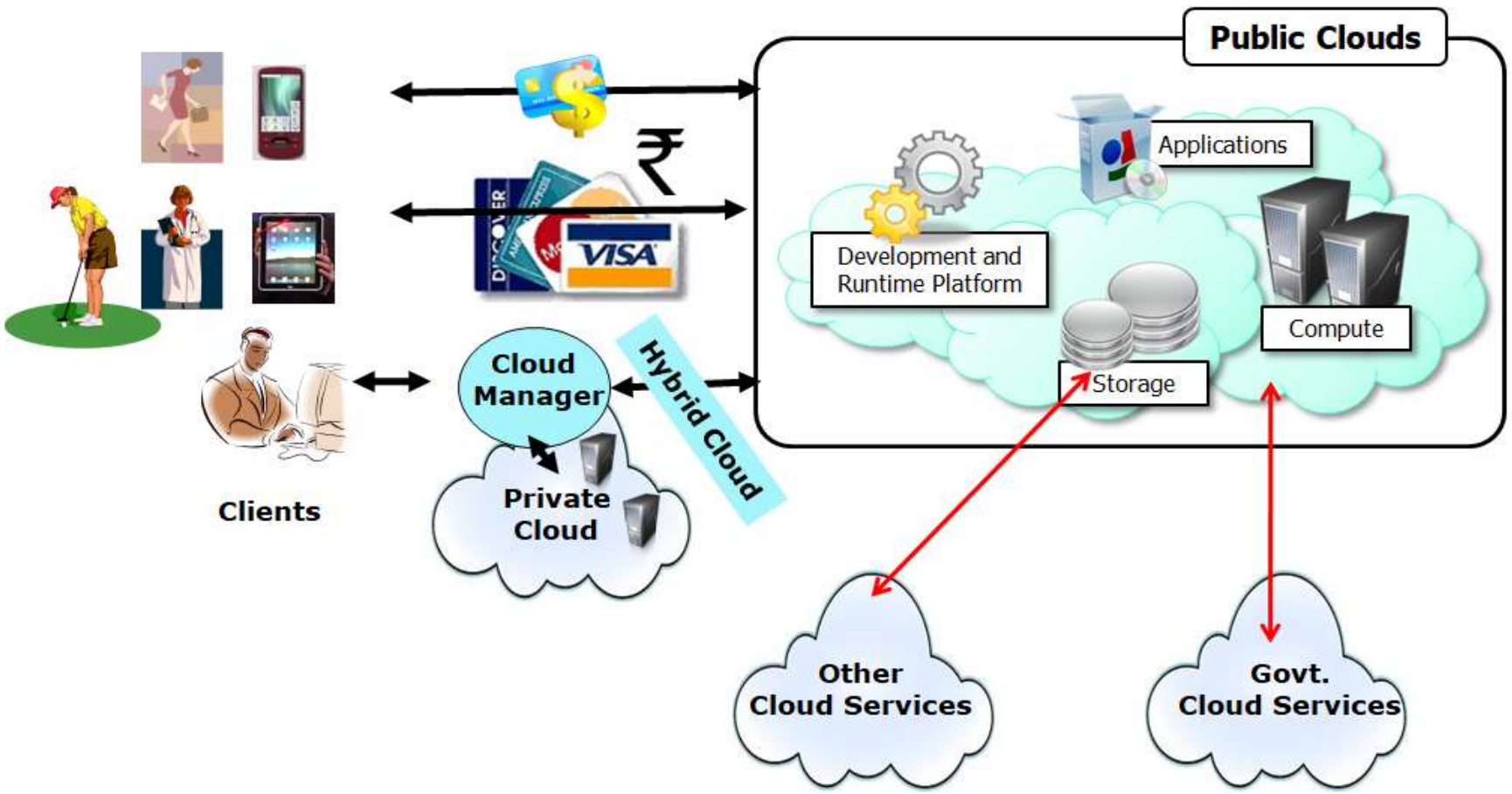
- **System developers can concentrate on the business logic rather than dealing with the complexity of infrastructure management and scalability**:
  - **Little Fluffy Toys is a company** in London that has developed a widget providing users with information about nearby rental bicycle services.
  - The company has managed to back the widget's computing needs on **Google AppEngine** and be on market in only one week.

A Closer Look in CC

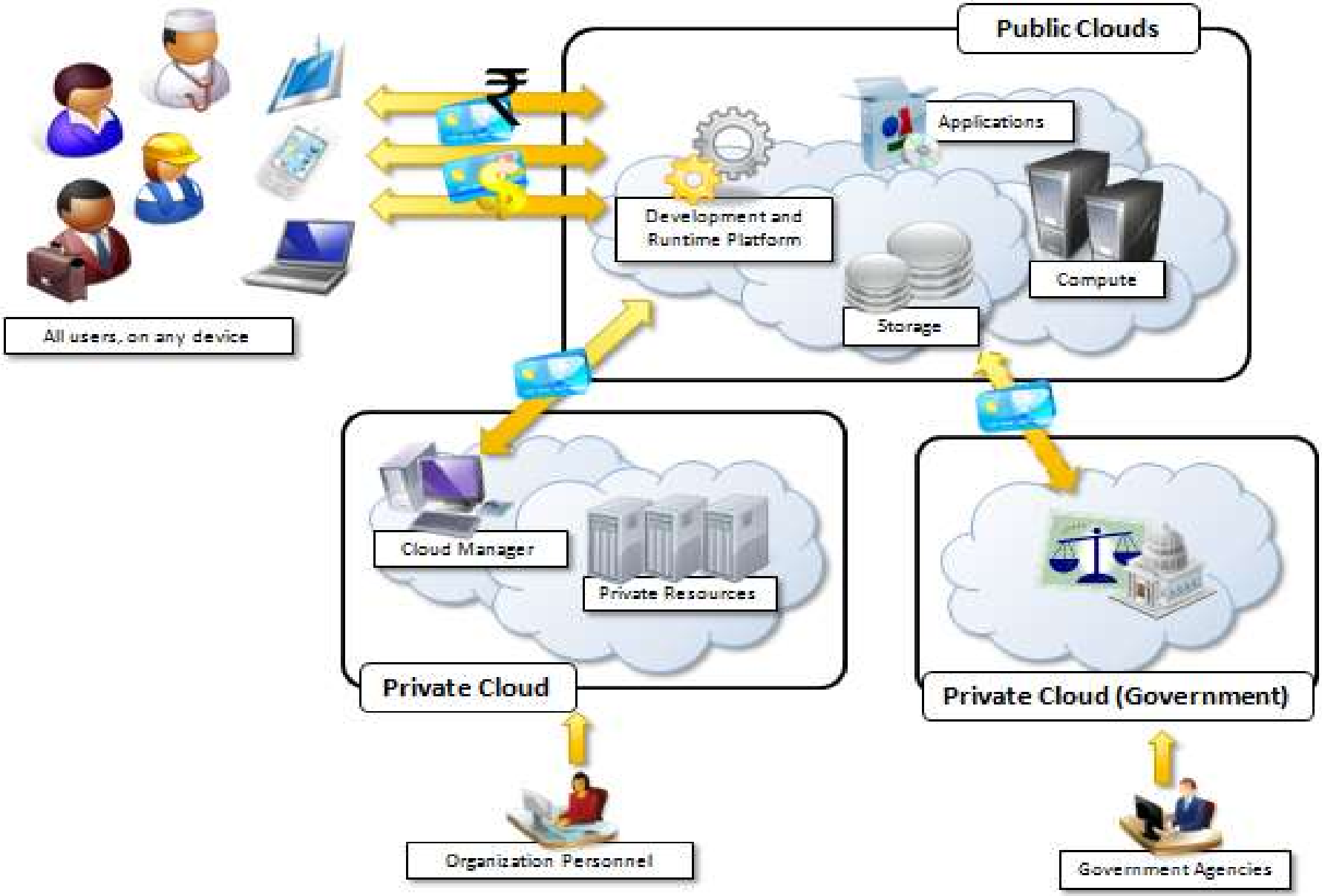
- **End users can have their documents accessible from everywhere and any device.**
  - **Apple iCloud** is a service that allows users to have their documents stored in the Cloud and access them from any device they connect to it.
  - This makes it possible taking a picture with a smart phone, going back home and editing the same picture on your laptop, and having it shown updated on their tablet.
  - This process is completely transparent to the users who do not have to set up cables and connect these devices with each other.

A Closer Look in CC

Cloud computing does not only contribute with the opportunity of easily accessing IT services on demand, but also introduces a new thinking about how IT services and resources should be perceived: as utilities. A bird eye view of Cloud computing environment is shown in Fig.

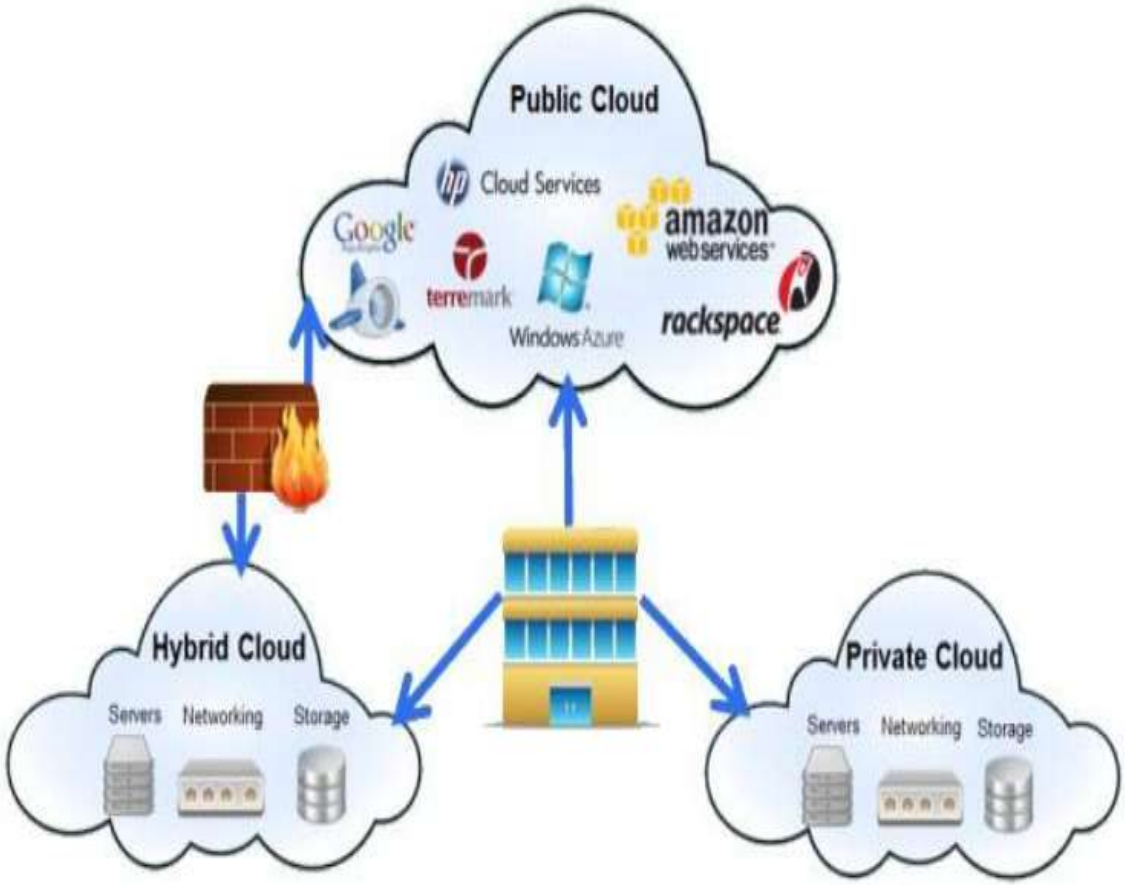


A Closer Look in CC



Major models for deployment

- Cloud Deployment Model functions as a virtual computing environment with a deployment architecture that varies depending on the amount of data you want to store and who has access to the infrastructure.



Major models for deployment

Public/Internet Clouds

* 3rd party, multi-tenant Cloud infrastructure & services:

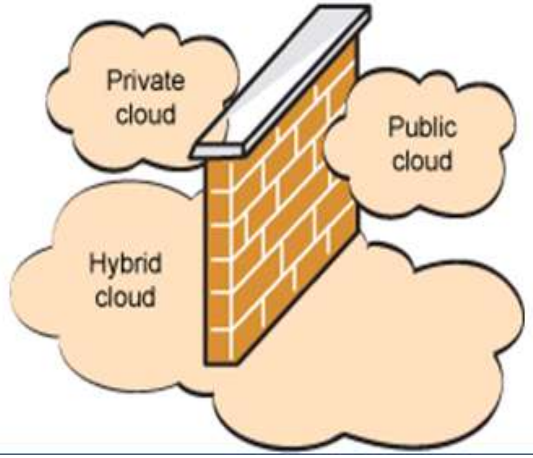
* available on subscription basis to all.

Private/Enterprise Clouds

* A public Cloud model within a company's own Data Center / infrastructure for internal and/or partners use.

Hybrid/Inter Clouds

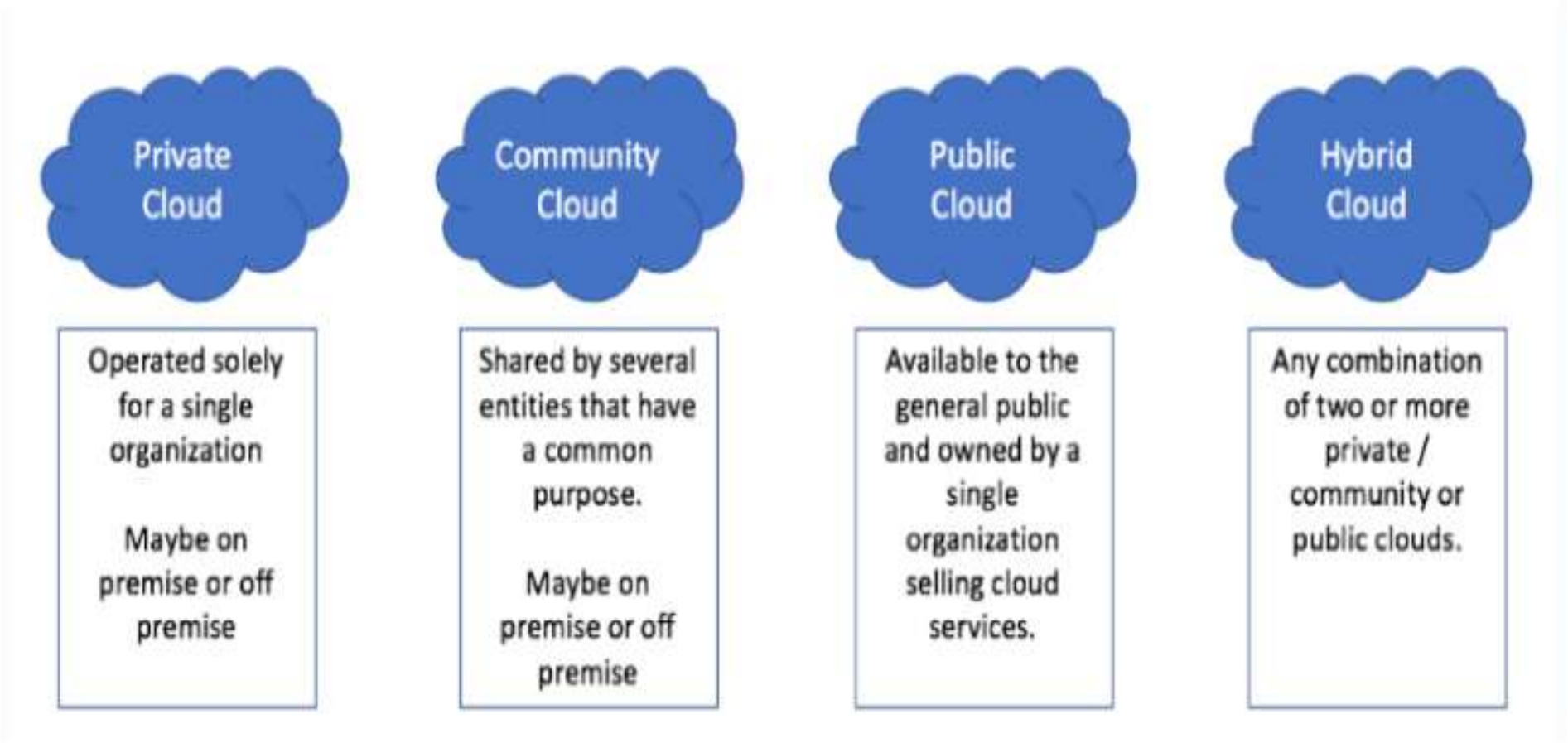
* Mixed usage of private and public Clouds: Leasing public cloud services when private cloud capacity is insufficient



Major models for deployment

- The three major models for deployment and accessibility of Cloud computing environments are:
  - public Clouds,
  - private/enterprise Cloud, and
  - hybrid Clouds
- **Public Cloud:** The public cloud makes it possible **for anybody to access** systems and services. The public cloud may be less secure as it is **open to everyone.**
- **Private Cloud:** The cloud computing resources that are **exclusively used inside a single business or organization** are termed as a private cloud. A private cloud may physically be located on the company's **on-site data centre** or hosted by a third-party service provider.
- **Hybrid Cloud:** It is the **combination of public and private clouds**, which is bounded together by technology that allows data applications to be shared between them. Hybrid cloud provides flexibility and more deployment options to the business.

Cloud Computing Deployment Models



Cloud-Computing Reference Model

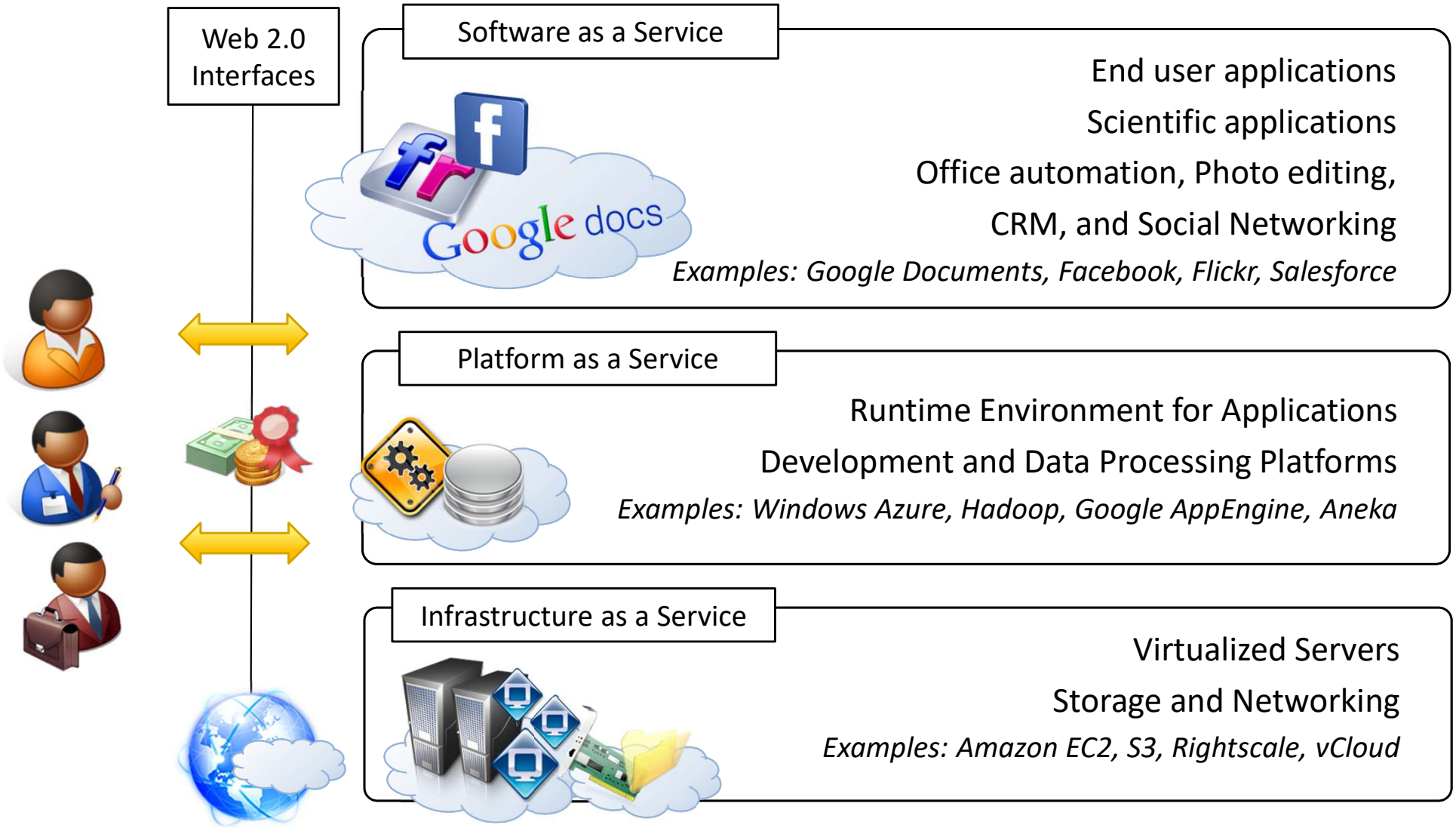
- A fundamental characteristic of Cloud computing is the capability of delivering on demand a variety of IT services, which are quite diverse from each other.

- This variety creates a different perception of what Cloud computing is among users. Despite this, it is possible to classify Cloud computing services offerings into three major categories:

  - *Infrastructure-as-a-Service (IaaS)*,
  - *Platform-as-a-Service (PaaS)*, and
  - *Software-as-a-Service (SaaS)*.

- These categories are related to each other as described in Fig, which provides an organic view of Cloud computing.

Cloud-Computing Reference Model



Cloud-Computing Reference Model

- **1. Infrastructure as a Service (IaaS):** It is a service where infrastructure is provided as outsourcing to enterprises such as networking equipment, devices, database, and web servers. It is also known as **Hardware as a Service (HaaS).**
  - **Flexibility and Control:** IaaS comes up with providing virtualized computing resources such as VMs, Storage, and networks facilitating users with control over the Operating system and applications.
  - **Reducing Expenses of Hardware**: IaaS provides business cost savings with the elimination of physical infrastructure investments making it cost-effective.
  - **Scalability of Resources:** The cloud provides in scaling of hardware resources up or down as per demand facilitating optimal performance with cost efficiency.

Cloud-Computing Reference Model

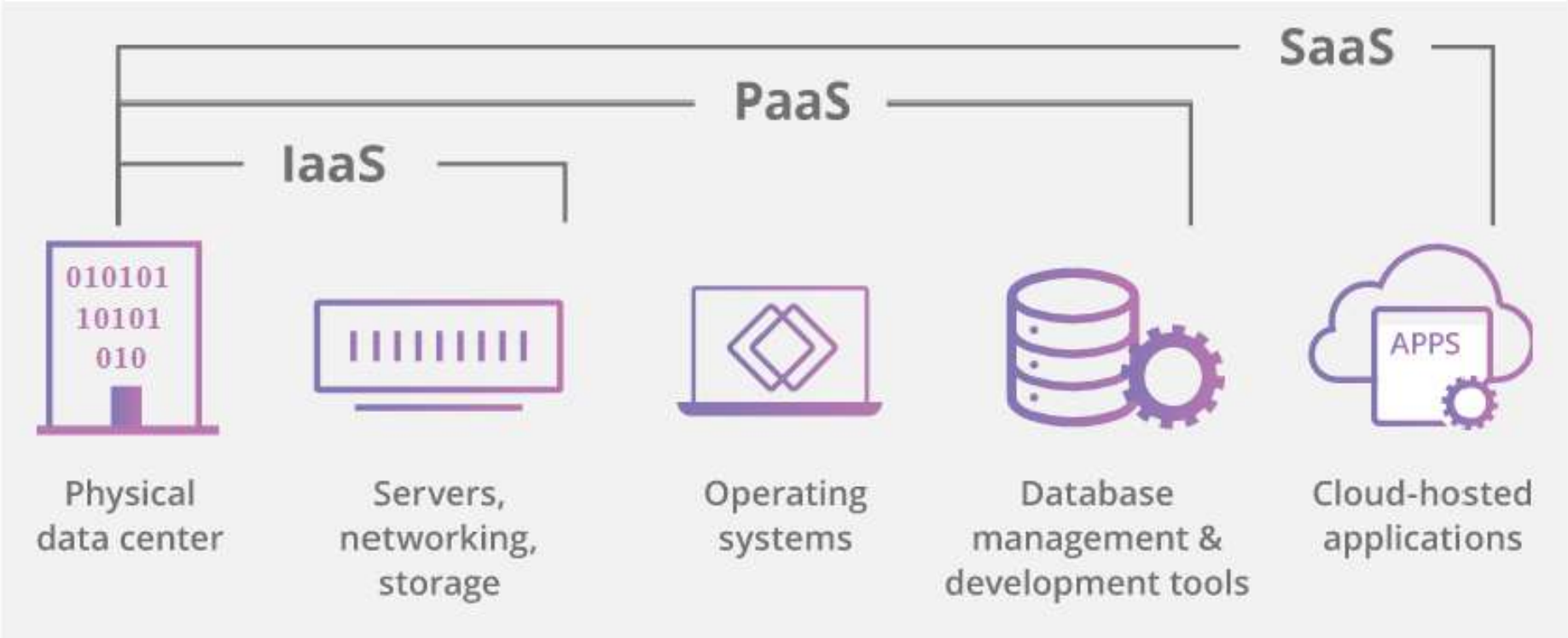
- **2. Platform as a Service (PaaS):** It is a category of cloud computing that provides a platform and environment to allow developers to build applications and services over the internet.
  - **Simplifying the Development:** Platform as a Service offers application development by keeping the underlying Infrastructure as an Abstraction. It helps the developers to completely focus on application logic (Code) and background operations are completely managed by the AWS platform.
  - **Enhancing Efficiency and Productivity:** PaaS lowers the Management of Infrastructure complexity, speeding up the Execution time and bringing the updates quickly to market by streamlining the development process.
  - **Automation of Scaling:** Management of resource scaling, guaranteeing the program's workload efficiency is ensured by PaaS.

Cloud-Computing Reference Model

- **3. SaaS (software as a service):** It is a way of delivering services and applications over the Internet. Instead of installing and maintaining software, we simply access it via the Internet, freeing ourselves from the complex software and hardware management.
  - **Collaboration And Accessibility:** Software as a Service (SaaS) helps users to easily access applications without having the requirement of local installations. It is fully managed by the AWS Software working as a service over the internet encouraging effortless cooperation and ease of access.
  - **Automation of Updates:** SaaS providers manage the handling of software maintenance with automatic latest updates ensuring users gain experience with the latest features and security patches.
  - **Cost Efficiency:** SaaS acts as a cost-effective solution by reducing the overhead of IT support by eliminating the need for individual software licenses.

Cloud-Computing Reference Model

- **Difference between IaaS, PaaS, and SaaS**



Characteristics and Benefits of CC

Characteristics

- Cloud computing has some interesting characteristics that bring benefits to both **Cloud Service Consumers(CSCs) and Cloud service providers (CSPs).**
- There are many characteristics of Cloud Computing here are few of them :
- **On-demand self-services:** The Cloud computing services ***does not require any human administrators***, user themselves are able to provision, monitor and manage computing resources as needed.
- **Broad network access:** The Computing services are generally provided over standard networks and heterogeneous devices.
- **Rapid elasticity:** The Computing services should have **IT resources** that are able to **scale out and in quickly and on a need basis**. Whenever the user require services it is provided to him and it is scale out as soon as its requirement gets over.
- **Resource pooling:** The IT resource (e.g., networks, servers, storage, applications, and services) present are **shared across multiple applications** and occupant in an uncommitted manner. Multiple clients are provided service from a same physical resource.
- **Measured service: The resource utilization is tracked** for each application and occupant, it will provide both the user and the resource provider with an account of what has been used. This is done for various reasons like monitoring billing and effective use of resource.

Characteristics and Benefits of CC

Characteristics

- **Multi-tenancy:** Cloud computing providers can support multiple tenants (users or organizations) on a **single set of shared resources**.
- **Virtualization:** Cloud computing providers use virtualization technology to abstract underlying hardware resources and present them as logical resources to users.
- **Flexible pricing models:** Cloud providers **offer a variety of pricing models**, including **pay-per-use, subscription-based, and spot pricing**, allowing users to choose the option that best suits their needs.
- **Security:** Cloud providers invest heavily in security measures **to protect their users' data** and ensure the **privacy of sensitive information**.
- **Automation:** Cloud computing services are often highly automated, allowing users to deploy and manage resources with **minimal manual intervention**.
- **Sustainability:** Cloud providers are increasingly focused on sustainable practices, such as energy-efficient data centers and the use of renewable energy sources, to reduce their environmental impact.

Characteristics and Benefits of CC

- **<u>Benefits</u>**
- **1) Agility**
- The cloud works in **a distributed computing environment**. It shares resources among users and **works very fast**.
- **2) High availability and reliability**
- The availability of servers is high and more reliable because the **chances of infrastructure failure are minimum**.
- **3) High Scalability**
- Cloud offers **"on-demand" provisioning of resources on a large scale**, without having engineers for peak loads.
- **4) Multi-Sharing**
- With the help of cloud computing, **multiple users and applications can work more efficiently** with cost reductions by sharing common infrastructure.

Characteristics and Benefits of CC

- <u>**Benefits**</u>
- **5) Device and Location Independence**
- Cloud computing enables the users to access systems using a web browser regardless of their location or what device they use e.g. PC, mobile phone, etc. **As infrastructure is off-site** (typically provided by a third-party) **and accessed via the Internet, users can connect from anywhere**.
- **6) Maintenance**
- Maintenance of cloud computing applications is easier, since they **do not need to be installed on each user's computer and can be accessed from different places**. So, it reduces the cost also.
- **7) Low Cost**
- By using cloud computing, the cost will be reduced because to take the services of cloud computing, **IT company need not to set its own infrastructure** and pay-as-per usage of resources.
- **8) Services in the pay-per-use mode**
- Application Programming Interfaces **(APIs) are provided to the users so that they can access services on the cloud** by using these APIs **and pay the charges as per the usage of services**.

Challenges Ahead of CC

- As any new technology develops and becomes popular, **new issues have to be faced**. Cloud computing is not an exception and new interesting problems and challenges are posed to the Cloud community, including IT practitioners, managers, governments, and regulators.
- So, if you are going to implement cloud computing, here are some of the challenges and risks you might face. I've also presented some strategies to prevent or minimize them.

Challenges Ahead of CC

- **1. Data Security and Privacy**
  - Data security is a **key concern** in cloud computing. Sensitive user or organizational data stored in the cloud requires **robust protection**. While **cloud providers ensure data integrity**, you must manage user **authentication, identity management, data encryption, and access control**.
  - Security risks like **identity theft, data breaches, and malware** can erode(damage) user trust, impacting revenue and reputation. Additionally, transferring large volumes of data at high speeds makes it vulnerable to leaks.
- **2. Cost Management**
  - Although cloud providers offer a **'Pay As You Go'** model **to reduce costs, enterprises can still face high expenses**. This happens due to under utilized resources, such as **servers not being fully used**, leading to hidden costs.
  - Degraded application performance, sudden usage spikes, and **unused resources** also contribute to increased expenses. For example, **forgetting to turn off cloud services during downtime** can result in unnecessary costs.

Challenges Ahead of CC

- **3. Multi-Cloud Environments**
  - Many enterprises now depend on multiple cloud service providers, **with 84% using hybrid cloud** strategies.
  - **Managing multiple clouds is often challenging** for infrastructure and IT teams due to differences **between providers, making the process complex.**
- **4. Performance Challenges**
  - Performance is crucial for cloud-based solutions. **Poor performance can drive users away and reduce profits**. Even slight delays in loading apps or web pages can lead to a significant drop in users.
  - This may result from **inefficient load balancing**, where the server fails to distribute traffic effectively. Fault tolerance is another challenge, ensuring operations continue despite component failures.
- **5. Interoperability and Flexibility**
  - **Switching between cloud providers can be tedious(**too long**)**. Applications often need to be rewritten for the new cloud's stack.
  - Complexities like **data migration, reconfiguring security, and setting up networks reduce flexibility** and make the process challenging.

Challenges Ahead of CC

- **6. High Dependence on Network**
  - Cloud computing relies on **real-time resource provisioning and large data transfers, enabled by high-speed networks**. However, **limited bandwidth or sudden outages can make it vulnerable**.
  - While enterprises **save on hardware costs, they must ensure sufficient bandwidth and no outages to avoid business losses**. This is especially challenging for smaller enterprises due to the high cost of maintaining network bandwidth.
- **7. Lack of Knowledge and Expertise**
  - Working with the cloud **is complex and requires extensive knowledge and expertise**. Professionals in this field must continually update their skills.
  - Despite **high demand and pay, there is a shortage of skilled cloud engineers and developers**. Up skilling is essential to help professionals effectively manage and develop reliable cloud-based applications.

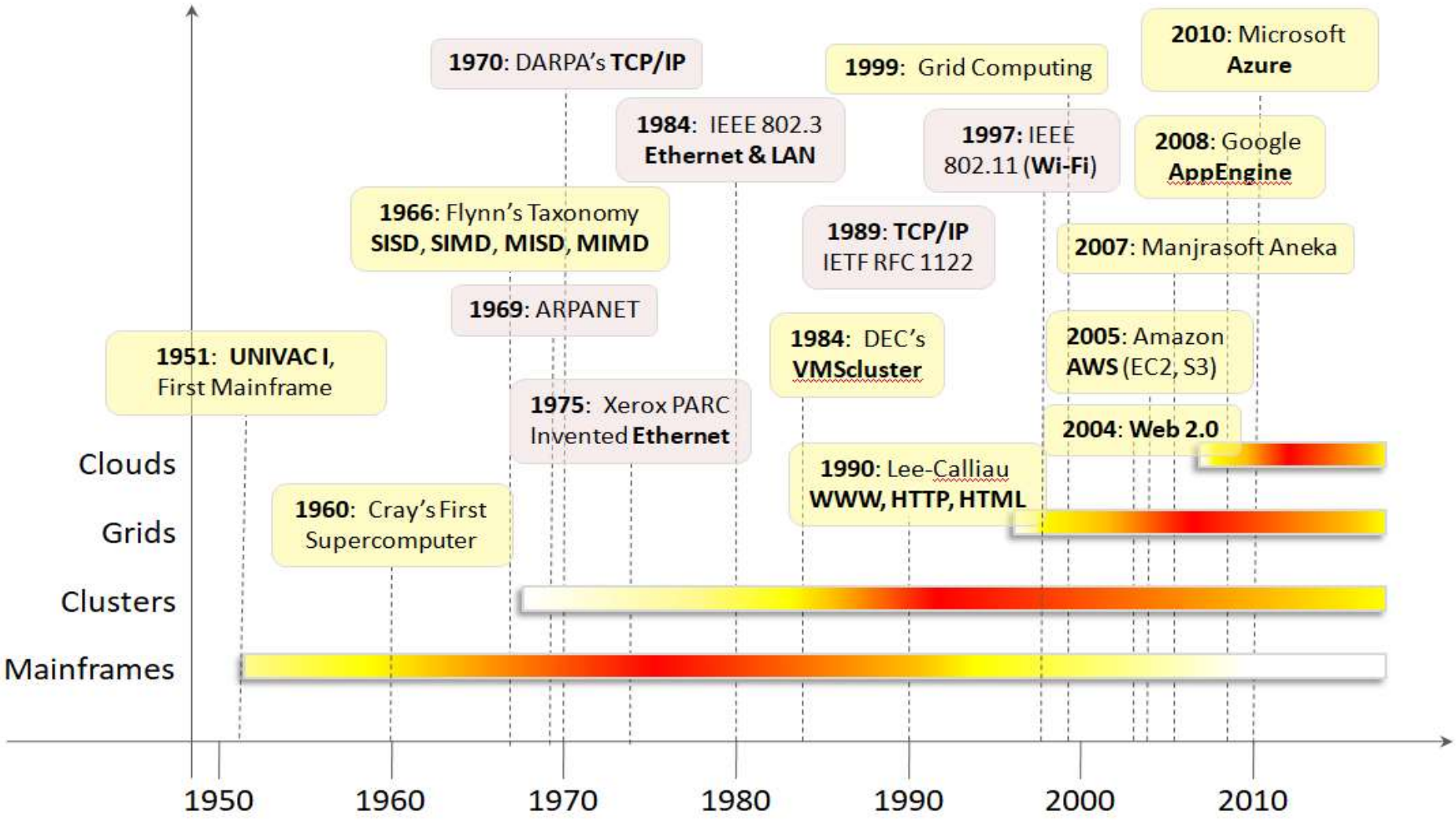
HISTORICAL DEVELOPMENTS

- The idea **of renting computing services** by large distributed computing facilities has been around for a long time.
- It dates back to the days of the **mainframes** in the **early fifties**. From there on, technology has evolved and refined.
- This process has created a series of favorable conditions for the realization of Cloud computing.

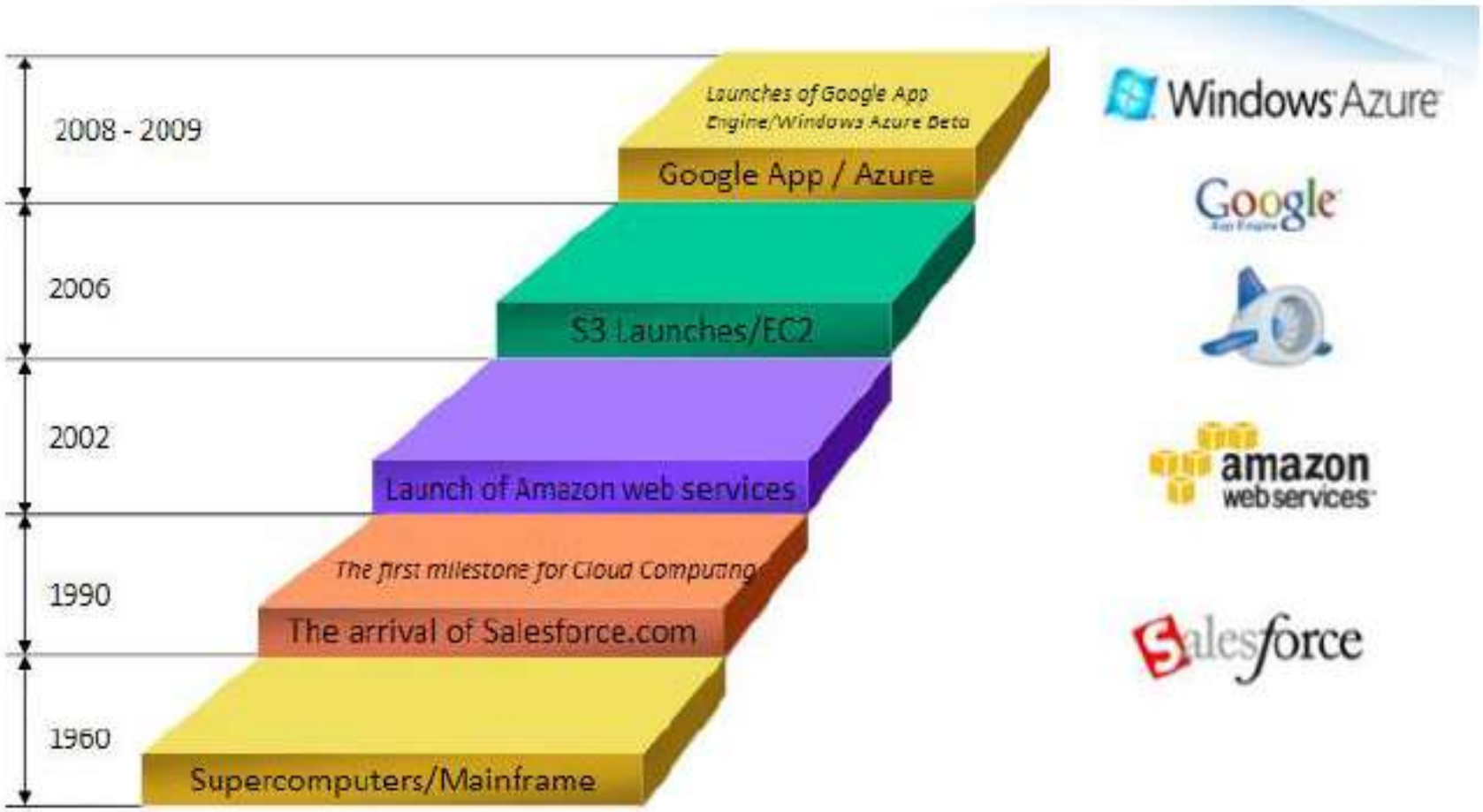
HISTORICAL DEVELOPMENTS

- In tracking the historical evolution, we briefly review **five core technologies** that played an important role in the realization of Cloud computing.
- These are:
  - Distributed systems,
  - Virtualization,
  - Web 2.0,
  - Service-oriented computing and
  - Utility-Oriented Computing
- Below Figure provides an overview of the evolution of the technologies for distributed computing that have influenced Cloud computing.

HISTORICAL DEVELOPMENTS

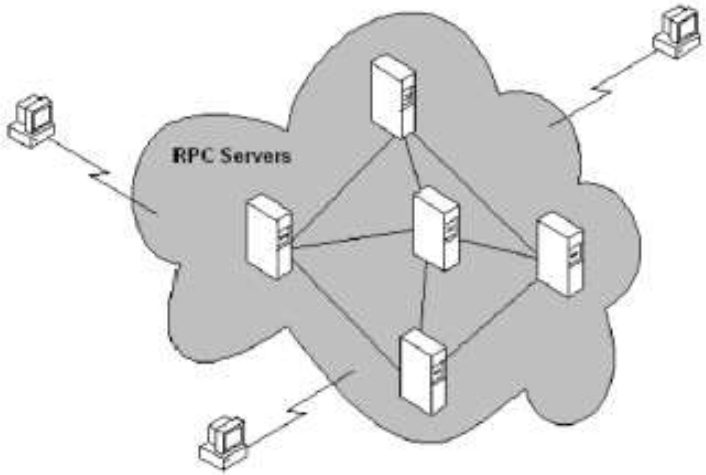


HISTORICAL DEVELOPMENTS



Distributed Systems

- Clouds are **essentially large distributed computing** facilities that make available their services to third parties on demand.
- *"A distributed system is a collection of independent computers that appears to its users as a single coherent system."*
- it evidences two very important elements characterizing a distributed system: the fact
  - it is composed of **multiple independent components** and that these components are perceived as a **single entity by users**.
  - The primary purpose of distributed systems is to **share resources and to utilize** them better.



Distributed Systems

- Distributed systems often exhibit other properties such as
  - *Heterogeneity(***dissimilar***)*,
  - *openness*,
  - *scalability*,
  - *transparency*,
  - *concurrency*,
  - *continuous availability*, and
  - *independent failures*.
- To some extent, these also characterize Clouds, especially in the
  - context of scalability,
  - concurrency, and
  - continuous availability.

Milestones of Cloud Computing

- Three major milestones have led to Cloud computing:
  - Mainframe computing,
  - Cluster computing, and
  - Grid computing.

a. Mainframes

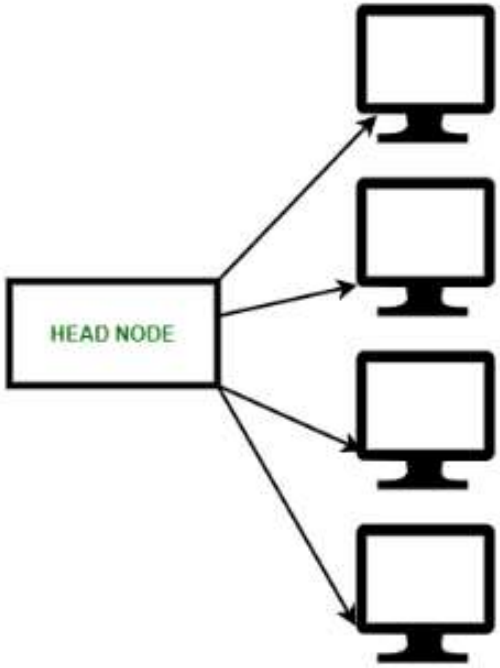
- These were the First examples of **large computational facilities leveraging multiple processing units**.
- Mainframes were **powerful, highly reliable** computers specialized for **large data movement and massive IO operations**.
- Used by large organizations for **bulk data processing** such as online transactions, enterprise resource planning, and other operations involving the processing of significant amount of data.

a. Mainframes

- The most attractive features of mainframes **is the ability to be highly reliable computers** that are “always on” and capable of tolerating failures transparently.
- System **shut down is not required to replace failed components**, and the system could work without interruptions.
- **Batch processing** is the main application of mainframes.
- **Popularity and deployments** of mainframes have **reduced**, but evolved versions of such systems are still in use for transaction processing (i.e., online banking, airline ticket booking, supermarket and telcos, and government services).

b. Clusters

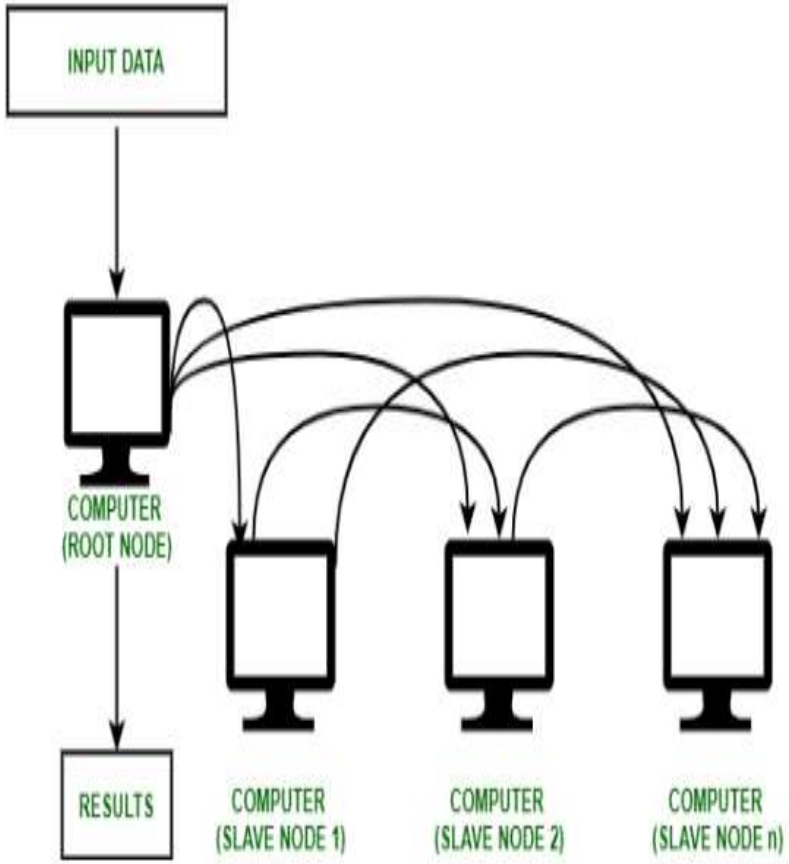
- Cluster computing is a collection of tightly (dependent) or loosely(independent) connected computers that work together so that they act as a single entity.
- The connected computers execute operations all together thus creating the idea of a single system. The clusters are generally connected through fast local_area networks (LANs)



Cluster Computing

Why is Cluster Computing important?

- Cluster computing gives a relatively **inexpensive, unconventional** to the large server or mainframe computer solutions.
- It resolves the demand **for content criticality and process services in a faster way**.
- Many organizations and IT companies are implementing cluster computing to augment their **scalability, availability, processing speed and resource management at economic prices.**
- It ensures that **computational power is always available**.
- It provides a single general strategy for the implementation and application of parallel high-performance systems independent of certain hardware vendors and their product decisions.



A Simple Cluster Computing Layout

b. Clusters

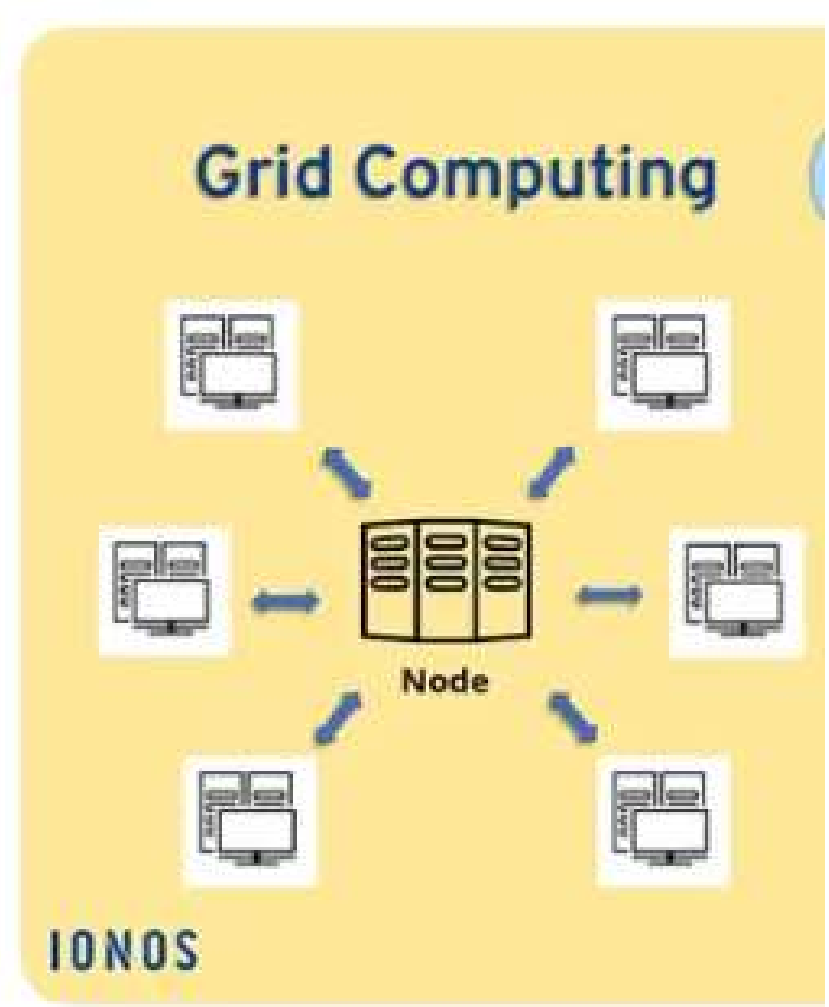
- Started as a **low-cost alternative** to the use of mainframes and supercomputers.
- **Computational power** of commodity machines could be leveraged to **solve problems previously manageable** only on expensive supercomputers.
- Cheaper computing machines could be connected by a **high-bandwidth network and controlled** by specific software tools that manage them as a single system.
- Started in **1980s,** clusters became the standard technology for **parallel and high-performance computing**.
- Cluster technology considerably contributed to the evolution of tools and framework for distributed computing i.e**. *Parallel Virtual Machine (PVM), and Message Passing Interface (MPI).***

c. Grids

- **Grid computing** is a distributed architecture that combines computer resources from different locations to achieve a common goal. It breaks down tasks into smaller subtasks, allowing concurrent processing.
- All **unused resources on multiple computers are pooled together** and made available for a single task. Organizations use grid computing to perform large tasks or solve complex problems that are difficult to do on a single computer.
- **Grid Computing** can be defined as a network of computers working together to perform a task that would rather be difficult for a single machine. All machines on that network work under the same protocol to act as a virtual supercomputer.

Working of Grid Computing

- A Grid computing network mainly consists of these three types of machines
- **Control Node:** A computer, usually a server or a group of servers which administrates the whole network and keeps the account of the resources in the network pool.
- **Provider:** The computer contributes its resources to the network resource pool.
- **User:** The computer that uses the resources on the network.



c. Grids

- Grid computing appeared in the **early 90s** as an evolution of cluster computing. Grid computing proposed a new approach to access large computational power, **huge storage facilities, and a variety of services.**
- Users **can "consume" resources** in the same way as they use other utilities such as power, gas, and water. Grids initially developed as aggregation of geographically dispersed clusters by means of Internet connection.
- These clusters belonged to **different organizations and arrangements** were made among them to **share the computational power.**
- Different from a "large cluster", a computing grid was a **dynamic aggregation** of heterogeneous computing nodes, and its scale was nationwide or even worldwide.

c. Grids

- Several reasons made possible the diffusion of computing grids:
  - i) clusters were now resources quite common;
  - ii) they were often under-utilized;
  - iii) new problems were requiring computational power going beyond the capability of single clusters;
  - iv) the improvements in networking and the diffusion of Internet made possible long distance high bandwidth connectivity

Virtualization

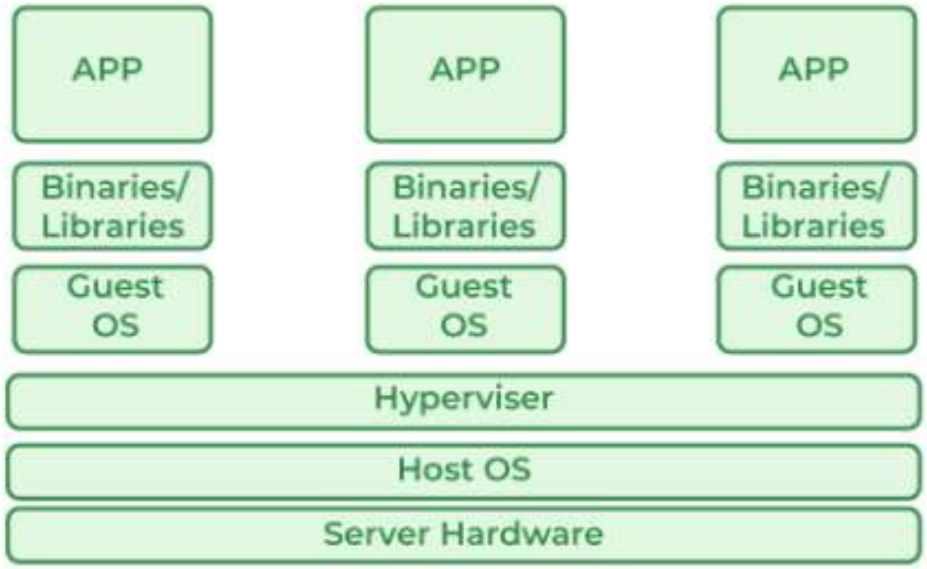
- Virtualization is another **core technology** for Cloud computing. It creates **simulated environments**, or virtual machines (VMs), from a single physical computer or server.
- This allows users to **run multiple operating systems** or applications on the **same physical machine**, while sharing resources. It was initially developed during the mainframe era.

Virtualization

- The most common example of virtualization is ***hardware virtualization***. This technology allows simulating the hardware interface expected by an **operating system**.
- This is the base technology that enables Cloud computing solutions delivering virtual server on demands, such as Amazon EC2, RightScale, VMware vCloud, and others.

Virtualization

- Creation of a virtual machine over existing operating system and hardware is known as Hardware Virtualization. A Virtual machine provides an environment that is logically separated from the underlying hardware.
- The machine on which the virtual machine is going to create is known as **Host Machine** and that virtual machine is referred as a **Guest Machine**



Virtualization

Virtualization

Benefits

- More flexible and efficient allocation of resources.
- Enhance development productivity.
- It lowers the cost of IT infrastructure.
- Remote access and rapid scalability.
- High availability and disaster recovery.
- Pay peruse of the IT infrastructure on demand.
- Enables running multiple operating systems.

Types of Virtualization

- Application Virtualization
- Network Virtualization
- Desktop Virtualization
- Storage Virtualization
- Server Virtualization
- Data virtualization

Web 2.0

- The Web is the **primary interface** through which **Cloud computing deliver its services**. At present time,
- it encompasses(cover) a set of technologies and services that facilitate interactive **information sharing, collaboration, user-centered design, and application composition**.
- This has **transformed the Web** into a **rich platform** for application development. Such evolution is known as "**Web 2.0**". This term captures a new way in which **developers architect applications, deliver services through the Internet, and provide a new user experience for their users**.

Web 2.0

- Web 2.0 revolutionizes web pages by making them **interactive and flexible**, similar to desktop applications. This is **achieved** through technologies like **XML, AJAX, and Web Services**, which allow users to contribute content.
- The widespread use of the internet opens new markets, and Web 2.0 applications are **dynamic, continuously improving** without the need for client-side updates.
- They support **lightweight deployment and programming models,** and new applications can be created by integrating existing services. This makes it easier to cater(provide) to user interests and make applications accessible and affordable for everyone.

Web 2.0

- Examples of Web 2.0 applications include Google Documents, Google Maps, Flickr, Facebook, Twitter, YouTube, de.li.cious, Blogger, and Wikipedia.
- Social networking sites like Facebook and Flickr benefit the most from Web 2.0, offering high levels of interaction to technologies like AJAX and RSS.
- These community websites rely on user-generated content, with platforms like Flickr for digital media storage, Facebook for social networking, and Blogger for online diaries.

Service-Oriented Computing

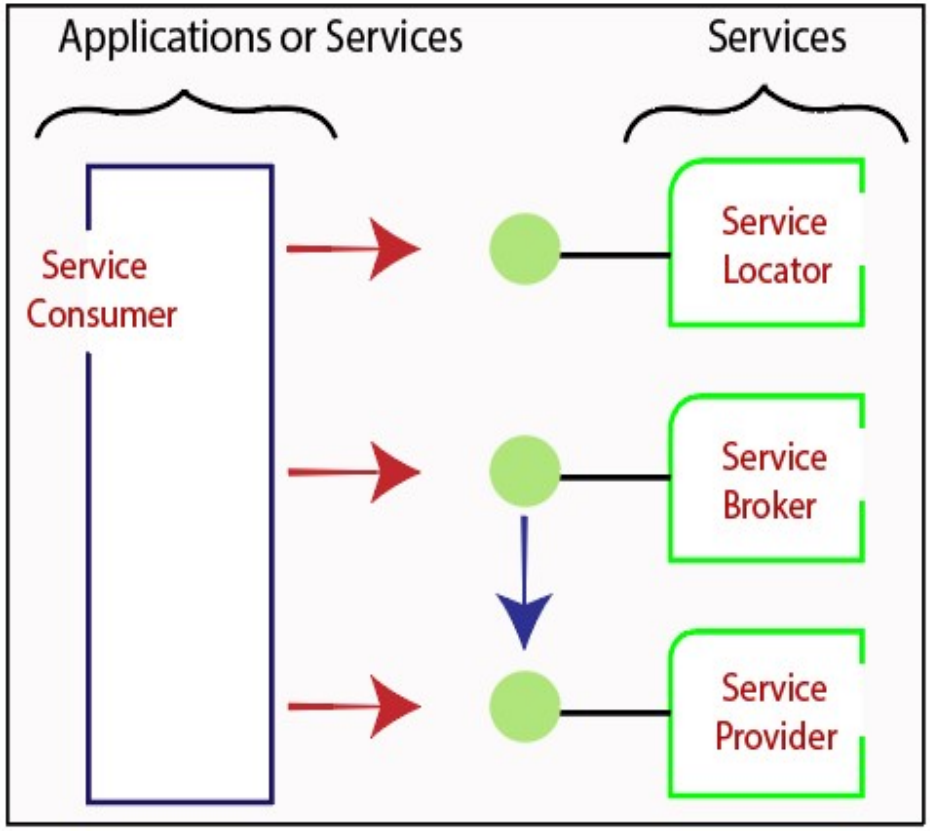
- Service orientation is the **core reference model** for Cloud computing systems. This approach adopts the concept of services as **main building blocks of application and system development**.
- *Service-Oriented Computing (SOC)* supports ***the development of rapid, low-cost, flexible, interoperable, and evolvable applications and systems.***
- A service is a **versatile(many different)** component that can perform any function, from **simple tasks to complex** business processes. It can be any piece of code that exposes its functionalities through a network-accessible protocol.

Service-Oriented Computing

- Services are designed to be
  - loosely coupled(reduces the interdependencies between components),
  - reusable,
  - independent of programming languages, and
  - location transparent.
- This means they can easily serve different scenarios, be **reused**, and **accessed** by a wide range of clients.
- Services are organized into a Service-Oriented Architecture (SOA), which allows them to be provided to end users or other entities over the network through published and discoverable interfaces.

Service-Oriented Architecture

- **Services -** The services are **the logical entities** defined by one or more published interfaces.
- **Service provider -** It is a software entity that implements a **service specification**.
- **Service consumer -** It can be called as a **requestor or client** that calls a service provider. A service consumer can be another service or an end-user application.
- **Service locator -** It is a service provider that acts as **a registry**. It is responsible for examining service provider interfaces and service locations.
- **Service broker -** It is a service provider that **pass service requests** to one or more additional service providers.



Service-Oriented Computing

- Service-Oriented Computing introduces two key concepts important for Cloud computing: Quality of Service (QoS) and Software as a Service (SaaS).
  - **Quality of Service (QoS)**: This measures a service's performance, including response **time, security, reliability, scalability, and availability.** QoS requirements are defined in a Service Level Agreement (SLA) between the client and provider.
  - **Software as a Service (SaaS)**: This model delivers applications over a network from a central data center on a subscription basis. The provider maintains the infrastructure, freeing the client from maintenance costs. SaaS uses multi-tenancy to offer individual software components separately, enabling complex business processes to be delivered as services and reused globally.

Utility-Oriented Computing

- Utility computing is a vision of **computing**, defining a service provisioning model for compute services in which **resources** such as **storage, compute power, applications, and infrastructure** are packaged and offered on a pay-per-use basis.
- The **idea** of **providing computing** as a *utility* **like natural gas, water, power, and telephone connection** has a long history but has become a reality today with the advent of Cloud computing.
- The concept of service provisioning **began with mainframes**, where companies like **IBM provided computing power to organizations**. This evolved with utility computing, which improved mainframe technology and introduced features like operating systems and user metering.

Utility-Oriented Computing

- The idea **extended** to academia with **cluster computing**, allowing research institutes to **use external IT infrastructure on demand**. The **internet and web technologies enabled utility computing on a global scale**, leading to the development of computing grids that offered distributed computing resources.
- **E-commerce** technologies further **supported utility computing** by making online services **like storage and web hosting accessible**. The dot-com bubble increased public interest in **buying online services**, leading to the development of reliable online payment systems.

Utility-Oriented Computing

- Service-oriented computing and Service-Oriented Architectures (SOAs) introduced the use of external services for **specific tasks within software systems**. Applications became a network of services from different providers, accessible via the Internet and charged based on usage.
- This **expanded the concept** of **utility computing** to include not just **compute power and storage**, but also services and application components. Quality of Service (QoS) became important to ensure reliable performance.
- These developments were crucial in the evolution of Cloud computing, where the vision of "computing utilities" is fully realized.

BUILDING CLOUD-COMPUTING ENVIRONMENTS

- The creation of Cloud-computing environments encompasses(cover) both the **development** of applications and **systems** that leverage(use) Cloud-computing solutions and the creation of **frameworks, platforms, and infrastructures** delivering Cloud-computing services.
- Here 2 types of Environments
  - Application Development
  - Infrastructure and System Development

Application Development

- Application Development refers to the process of creating computer programs or software applications that run on the Internet instead of on your personal computer or a local server.

Application Development

- Applications that leverage(use) Cloud-computing benefit from its capability of **dynamically scaling** on demand.
- **One class of applications** that take the **biggest advantage** from this feature is **Web applications**. Their performance is mostly influenced by the **workload generated** by varying user demands.
- With **Web 2.0 technologies**, the web has become a **platform for developing rich, complex applications**, including enterprise applications that use the internet for service delivery and user interaction.
- These applications involve **complex processes** triggered by **user interactions and multiple backend tiers**. They are sensitive to improper infrastructure **sizing, service deployment, or workload variability**.

Application Development

- **Another class** of application that benefits greatly from Cloud computing is **resource-intensive applications**. These can be either **data-intensive or compute-intensive.**
- They **require a lot of resources to complete tasks quickly**, but not constantly. For example, **scientific applications** may need huge computing power for large-scale experiments occasionally, making it impractical to buy the necessary infrastructure.
- Cloud computing provides a solution by **offering resources on demand**. These applications are usually **not interactive and involve batch processing**.

Application Development

- Cloud computing offers solutions for on-demand and dynamic scaling across all computing layers by:
  - Renting compute power, storage, and networking.
  - Providing scalable and dynamically sized runtime environments.
  - Offering application services that mimic desktop applications but are hosted and managed by the provider.
- These capabilities use service orientation for easy integration into existing systems.
- Developers access these services through simple web interfaces, often using REST web services, making cloud application development and management practical and straightforward.

Infrastructure and System Development

- Distributed computing, virtualization, service orientation, and Web 2.0 form the **core technologies** enabling the provisioning of Cloud services from **anywhere in the globe**.
- Developing applications and systems that leverage(use) the Cloud requires knowledge across all these technologies. Moreover, new challenges need to be addressed from design and development standpoints.
- **Distributed computing** is essential for Cloud computing because Cloud systems are **inherently distributed**.

Infrastructure and System Development

- The **main challenge for engineers and developers** is **managing the dynamic nature** of Cloud systems, where **new nodes and services are added** on demand. This is mainly addressed at the **middleware layer**.
- **Infrastructure-as-a-Service (IaaS)** solutions allow **adding and removing resources**, but effective use depends on the deployment.
- **Platform-as-a-Service (PaaS)** solutions include **algorithms and rules** for resource provisioning, which can be either **transparent to developers or controlled by them**. Integrating Cloud resources with existing systems is also a key concern.

Infrastructure and System Development

- **Web 2.0** technologies are the interface for **delivering, managing, and provisioning** Cloud computing services. **Web services are the main access point** for Cloud systems from a programming perspective.
- **Service orientation** is the key paradigm for Cloud architecture, often summarized as XaaS (everything as a service).
- Despite the lack of a single standard for accessing resources from different Cloud providers, the common technology makes it easier to learn and integrate Cloud computing into existing systems.

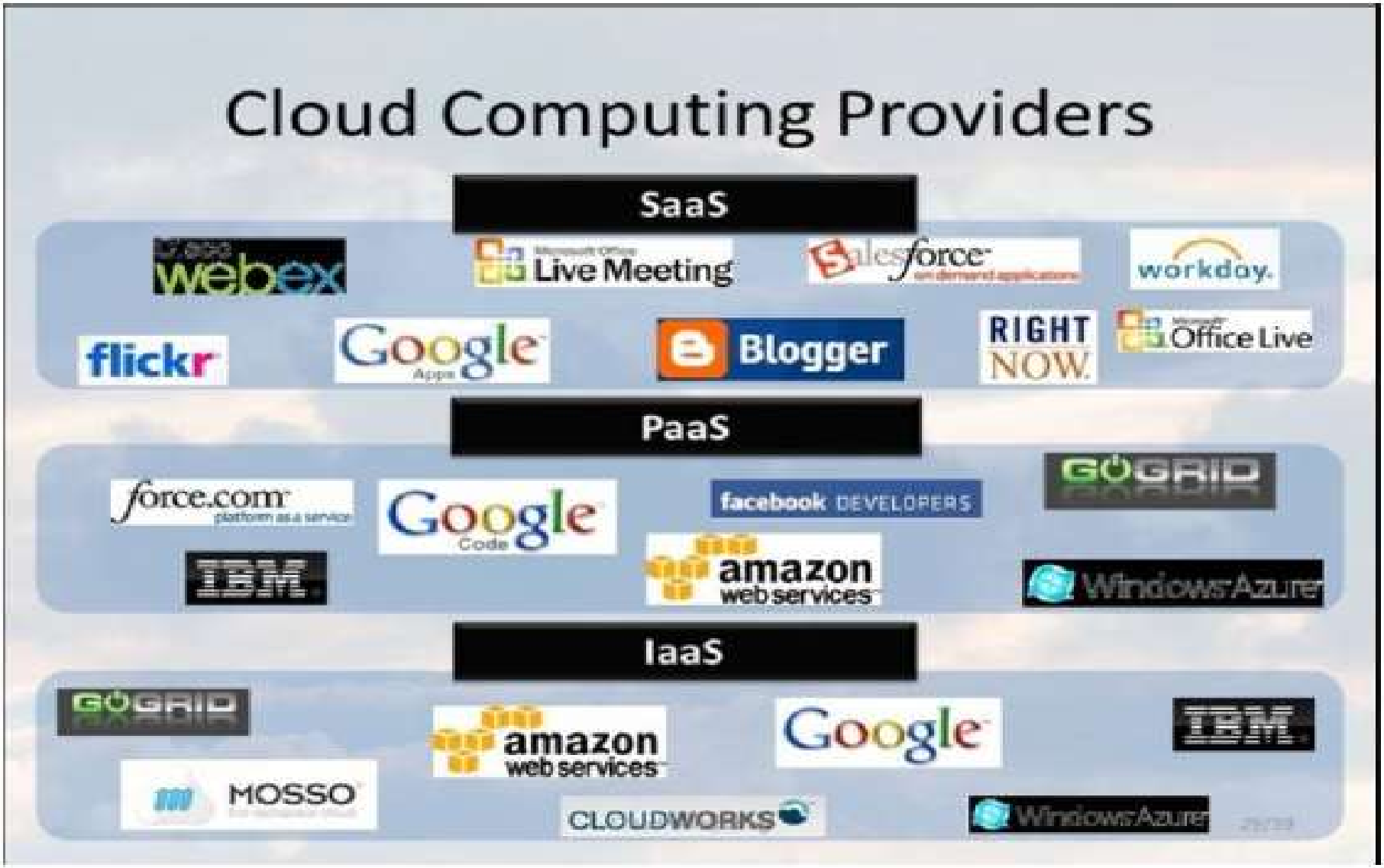
Infrastructure and System Development

- **Virtualization is crucial** for Cloud computing, forming the backbone of Cloud infrastructure. Although it's an old concept, Cloud computing **brings new challenges** in managing virtual environments, whether they involve **virtual hardware or runtime environments.**
- Developers must **understand the limitations and volatility** (change quickly)of their chosen virtualization technology. These factors influence how we **design and program** Cloud-based applications and systems.
- Cloud computing addresses demand surges by **replicating stressed components, with dynamism, scale, and volatility** being key design considerations.

COMPUTING PLATFORMS AND TECHNOLOGIES

- Development of a Cloud computing application happens by leveraging(use) platform and frameworks that provide different types of services, from the bare metal infrastructure to customizable applications serving specific purposes.
  - Amazon Web Services (AWS)
  - Google AppEngine,
  - Microsoft Azure,
  - Hadoop
  - Force.com and Salesforce.com,
  - Manjrasoft Aneka

COMPUTING PLATFORMS AND TECHNOLOGIES



Amazon Web Services (AWS)

- AWS offers comprehensive Cloud IaaS services, ranging from virtual compute, storage, and networking to complete computing stacks. AWS is mostly known for its compute and storage on demand services, namely ***Elastic Compute Cloud (EC2)*** **and** ***Simple Storage Service (S3)*****.**
- **EC2** offers customizable **virtual hardware** for deploying cloud computing systems. Users can choose from **various configurations**, including **GPU and cluster instances**. EC2 instances can be deployed using the AWS console or Web services API.
- **EC2** also allows **saving running instances** as images to create deployment templates, which are **stored in S3** for persistent storage. **S3 organizes data into buckets**, which can **store objects of any size**, from simple files to entire disk images.
- Additionally, AWS provides various services like networking support, caching systems, DNS, and database support to build virtual computing systems.

Amazon Web Services (AWS)

Google AppEngine

- Google AppEngine is a scalable runtime environment mostly devoted to executing Web applications. These take advantage of the large computing infrastructure of Google to **dynamically scale** as the demand varies over time.
- AppEngine offers a **secure environment and various services** to simplify developing scalable, high-performance web applications. These services include **in-memory caching, scalable data storage, job queues, messaging, and cron tasks**.
- Developers can build and test applications locally using the AppEngine SDK, which mimics the production environment. Once development is complete, applications can be easily migrated to AppEngine, cost quotas can be set, and the application can be made available globally. Supported languages include Python, Java, and Go.

Google AppEngine

Microsoft Azure

- Microsoft Azure is a cloud operating system and platform for developing cloud applications. It provides a scalable environment for web and distributed applications. Azure applications are organized into three types of roles:
  - **Web Role**: Hosts web applications.
  - **Worker Role**: A generic container for applications, used for workload processing.
  - **Virtual Machine Role**: Provides a customizable virtual environment, including the operating system.
- Azure also offers additional services like storage (relational data and blobs), networking, caching, and content delivery to support application execution.

Microsoft Azure

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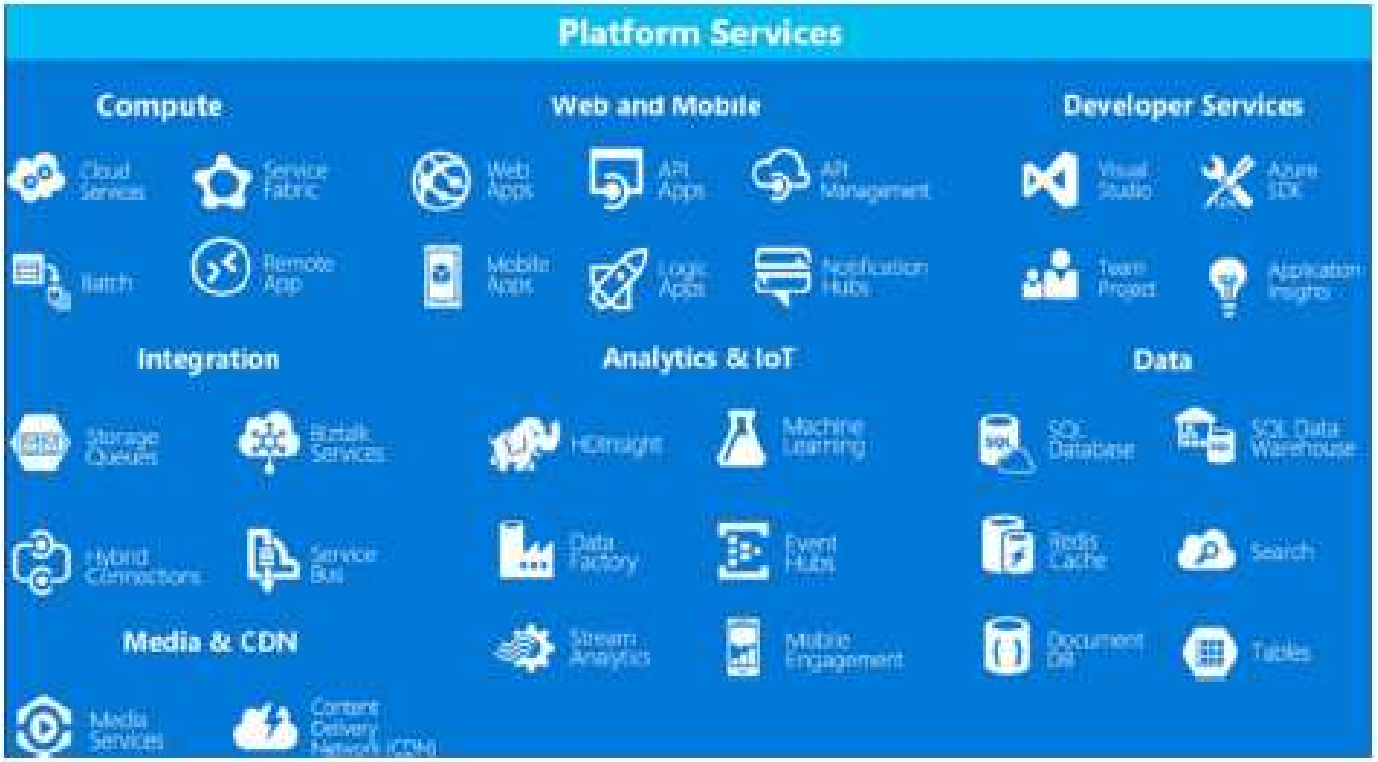
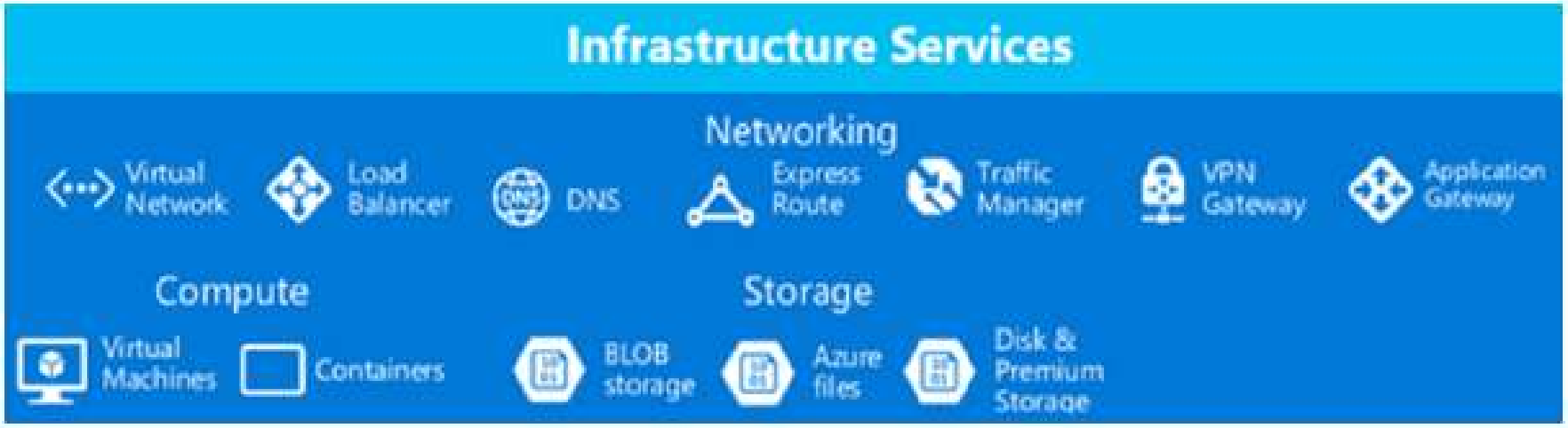
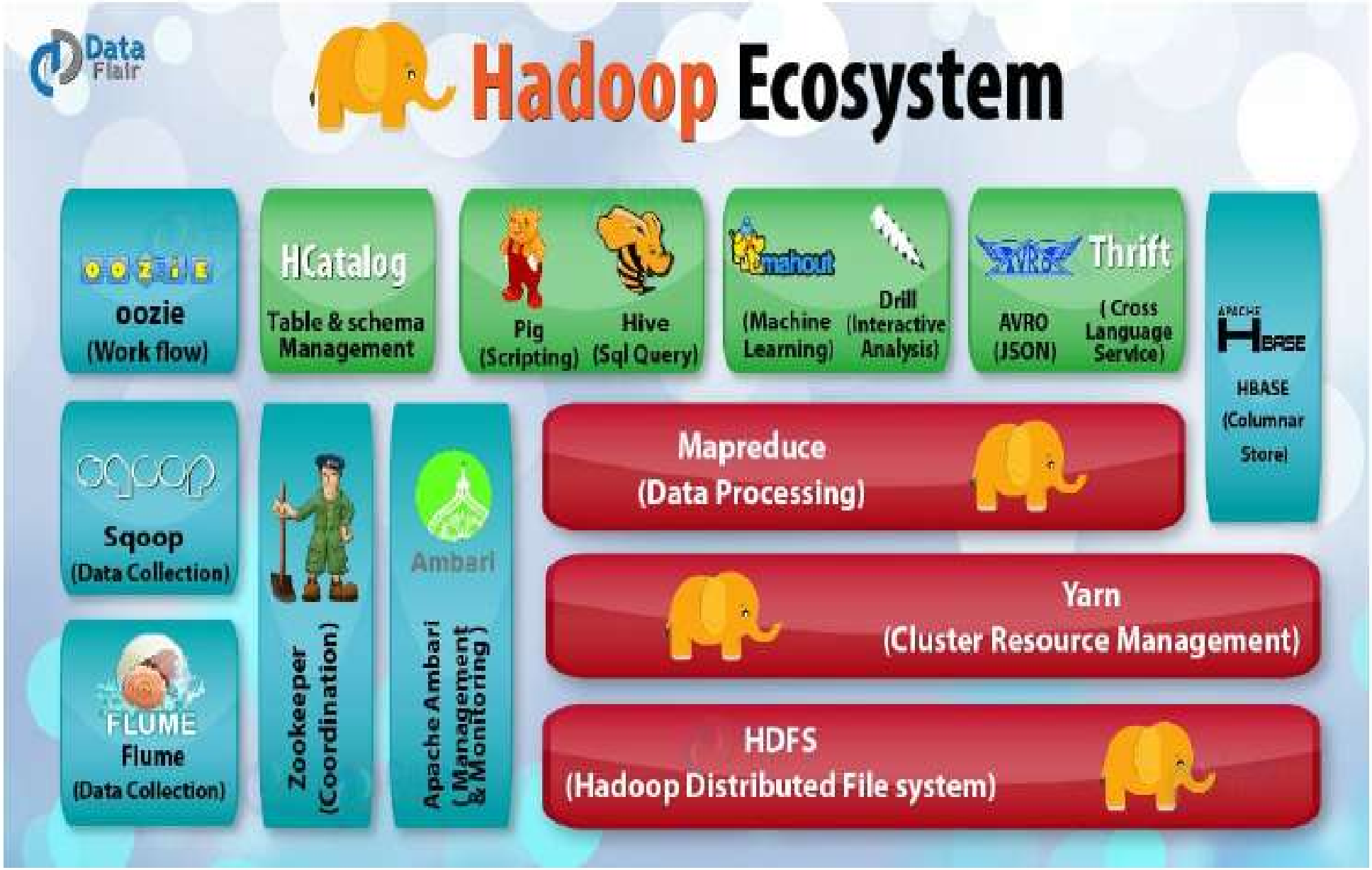




Hadoop

- Apache Hadoop is an open-source framework for **processing large data sets** on commodity hardware. It implements **MapReduce**, a model developed by Google, with **two key** operations: map (transforming input data) and reduce (aggregating output).
- Developers provide the **input data and specify the map and reduce functions**. Yahoo! sponsors Hadoop, making it an enterprise-ready cloud computing platform.
- Hadoop is **integral to Yahoo**!'s infrastructure and supports many business processes. Yahoo! also manages the world's largest Hadoop cluster, available to academic institutions.

Hadoop

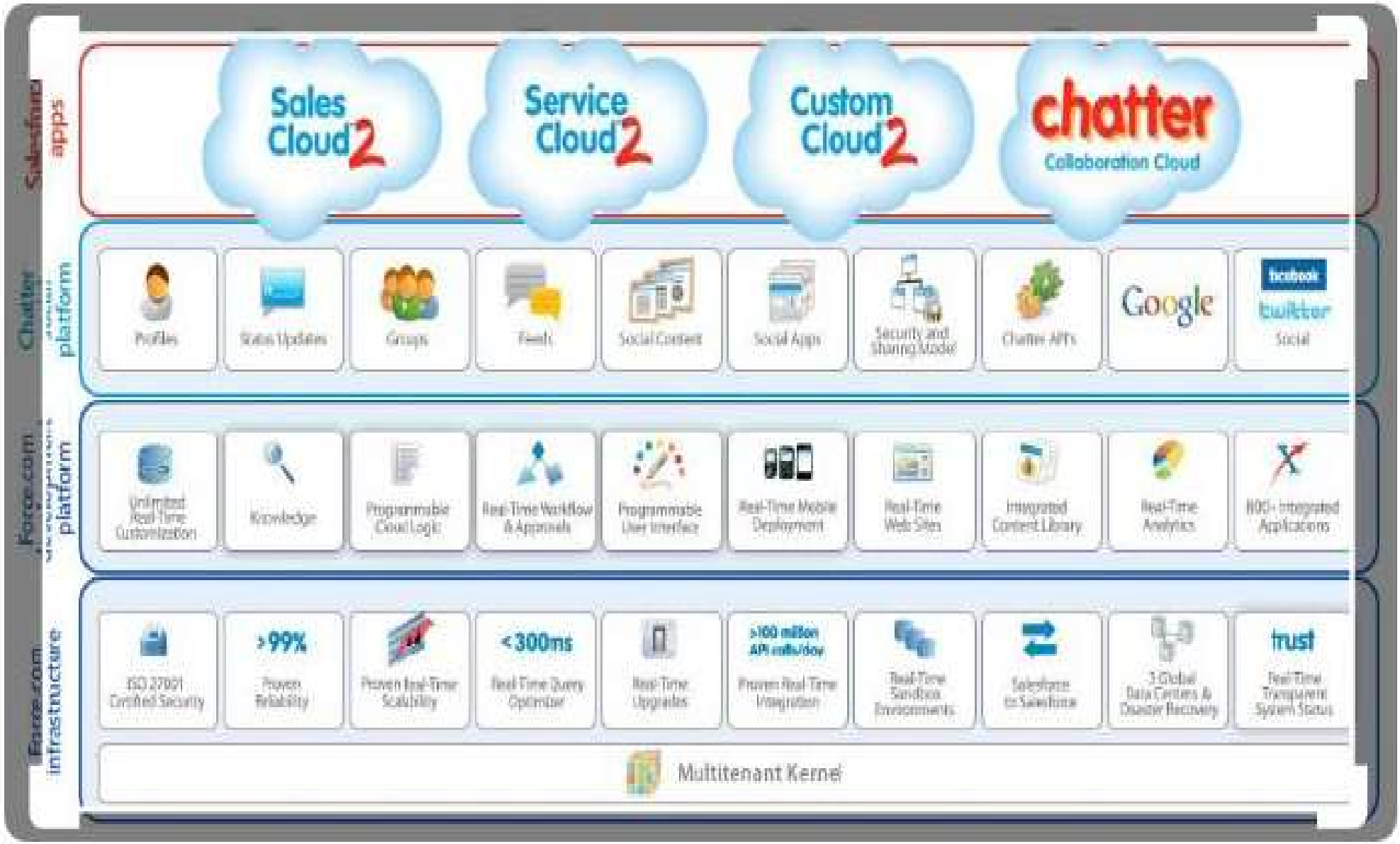

Data Flair
Hadoop Ecosystem
oozie
(Work flow)
HCatalog
Table & schema Management
Pig
(Scripting)
Hive
(Sql Query)
mahout
(Machine Learning)
Drill
(Interactive Analysis)
AVRO
(JSON)
Thrift
(Cross Language Service)
HBASE
(Columnar Store)
Sqoop
(Data Collection)
Flume
(Data Collection)
Zookeeper
(Coordination)
Ambari
Apache Ambari
(Management & Monitoring)
Mapreduce
(Data Processing)
Yarn
(Cluster Resource Management)
HDFS
(Hadoop Distributed File system)

Force.com and Salesforce.com

- Force.com is a cloud platform for **developing social enterprise applications** and is the foundation of SalesForce.com, **a SaaS** solution for customer relationship management. It allows **creating applications using ready-to-use** components or custom ones, and integrating AppExchange components.
- The platform supports all stages of application development, from data layout design to business rules, workflows, and user interface definition.
- Fully hosted on the cloud, Force.com provides complete access to its functionalities and those of hosted applications through web services technologies.

Force.com and Salesforce.com

Force.com and Salesforce.com



Manjrasoft Aneka

- Manjrasoft Aneka is a cloud application platform for **quickly creating scalable applications and deploying** them on various types of clouds seamlessly.
- It supports **different programming abstractions** for developing applications and a distributed runtime environment that can be deployed on **various hardware, including clusters, networked desktop computers, and cloud resources**.
- Developers can choose from **tasks, distributed threads, and map-reduce** to design their applications. These applications run on a **flexible, service-oriented runtime** environment that can dynamically add resources as needed.
- The runtime environment manages activities like **scheduling, execution, accounting, billing, storage, and quality of service**.

Manjrasoft Aneka

- Aneka is an example of cloud computing technology, These mostly **three major** market segments: **Infrastructure-as-a-Service, Platform-as-a-Service, and Software-as-a-Service**.
- It is used as a reference platform for discussing practical implementations of distributed applications, showing how clouds can be leveraged using various programming models and abstractions provided by Aneka.